

Plan of the course

- 1 How to describe knowledge and information?
- 2 What does rationality imply about behavior?
- 3 What is information worth? How should it be chosen?
- 4 What information matters for strategic predictions?

One decision maker $\longrightarrow$ several interacting players

Four lectures

1

Knowledge and agreement

Partitions; common knowledge; Aumann's theorem

2

Information and rational behavior

Minimax; correlated equilibrium; rationalizability

3

Comparing and choosing information

Blackwell; strategic value; information design

4

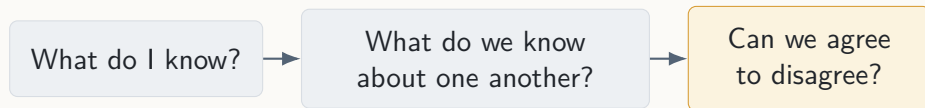
Higher-order beliefs and universal representation

Email game; universal type space; interim rationalizability

Advanced Game Theory

Knowledge and agreement

Lecture 1



Olivier Gossner

Lecture 1

- 01 Knowledge**
Possibilities and partitions
- 02 Common knowledge**
Reasoning about others
- 03 Agreement**
Common priors and posterior beliefs

1.1 Possibilities and partitions

Non-delusion and introspection

1.2 Knowledge operators

Deduction, truth and introspection

1.3 Linking the representations

Recover possibilities; match the axioms

The same weather; different observations

$\Omega = \{R, S\}$ R : rain S : sunshine

Weather observed



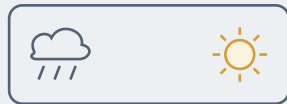
R



S

$\{\{R\}, \{S\}\}$

Weather unobserved



R

S

$\{\{R, S\}\}$

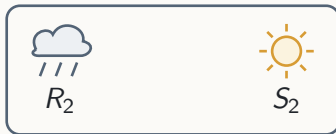
A state also describes information

$$\Omega = \{R_1, S_1, R_2, S_2\}$$

informed



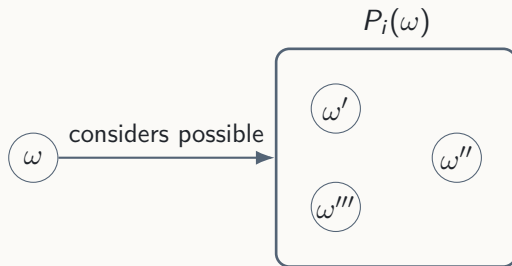
uninformed



Possibility correspondences

Finite state space Ω ; finite player set I ; player $i \in I$.

$$P_i : \Omega \longrightarrow 2^\Omega$$



Only the first digit is visible

$$\Omega = \{00, 01, \dots, 99\}$$

7

?

					⋮				
60	61	62	63	64	65	66	67	68	69
70	71	72	73	74	75	76	77	78	79
80	81	82	83	84	85	86	87	88	89
					⋮				

$$P_i(73) = ?$$

Axioms for possibility correspondences

Non-delusion

$$\omega \in P_i(\omega)$$

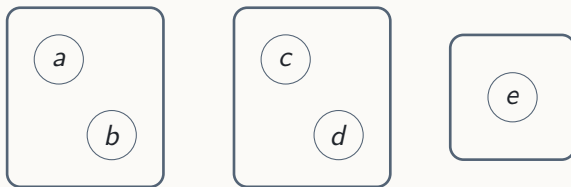
The truth remains possible.

Introspection

$$\omega' \in P_i(\omega)$$

$$\implies P_i(\omega') = P_i(\omega)$$

Why partitions?



Non-delusion and introspection $\iff$ partitional information

$P_i(\omega)$ = the cell containing ω

Why the cells form a partition

Assume non-delusion and introspection.

$$\omega' \in P_i(\omega) \iff P_i(\omega') = P_i(\omega)$$

Forward

Introspection.

Backward

$$\omega' \in P_i(\omega') = P_i(\omega)$$

Non-delusion at ω' .

States with the same possibility set form one cell.

	Non-delusion	Introspection
Reads the first digit only	?	?
Reads the digits in reverse	?	?
Never excludes the good state	?	?
Always assumes the good state	?	?

Write the possibility correspondence first.

Knowledge: Knowledge operators

1.1 Possibilities and partitions

Non-delusion and introspection

1.2 Knowledge operators

Deduction, truth and introspection

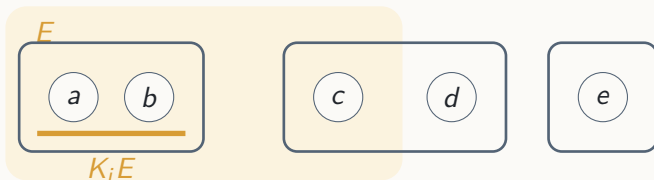
1.3 Linking the representations

Recover possibilities; match the axioms

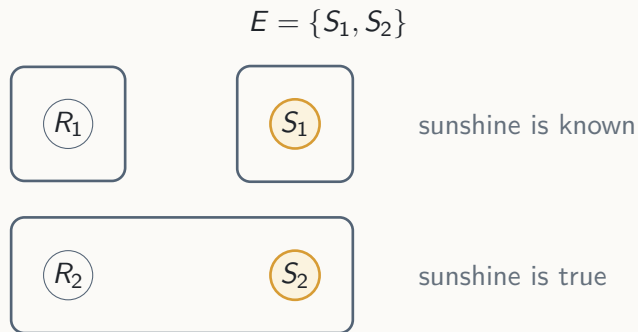
An event is known when every possibility lies inside it

An event is a subset $E \subseteq \Omega$.

$$K_i E = \{\omega \in \Omega : P_i(\omega) \subseteq E\}$$



Knowing that the sun shines



$$K_i E = \{S_1\} \subsetneq E$$

Axioms for knowledge: deduction and truth

Deductive closure

$$K_i(E \cap F) = K_iE \cap K_iF$$

Truth

$$K_iE \subseteq E$$

Axioms for knowledge: introspection

$$\neg E = \Omega \setminus E.$$

Positive introspection

$$K_i E \subseteq K_i K_i E$$

Negative introspection

$$\neg K_i E \subseteq K_i \neg K_i E$$

Both hold for partitional information.

Knowledge: Linking the representations

1.1 Possibilities and partitions

Non-delusion and introspection

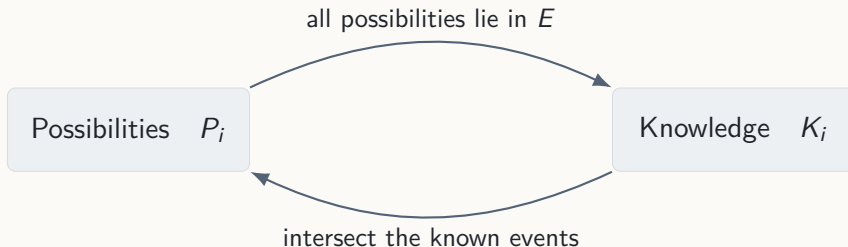
1.2 Knowledge operators

Deduction, truth and introspection

1.3 Linking the representations

Recover possibilities; match the axioms

Two descriptions of the same information



$$P_i(\omega) = \bigcap_{E: \omega \in K_i E} E$$

Finite Ω ; deductive closure and $K_i \Omega = \Omega$.

Possibilities and knowledge: the dictionary

Non-delusion $\iff$ Truth

Introspection $\iff$ Positive and negative introspection

$$K_i E = \{\omega : P_i(\omega) \subseteq E\}, \quad P_i(\omega) = \bigcap_{E: \omega \in K_i E} E$$

02

Common knowledge

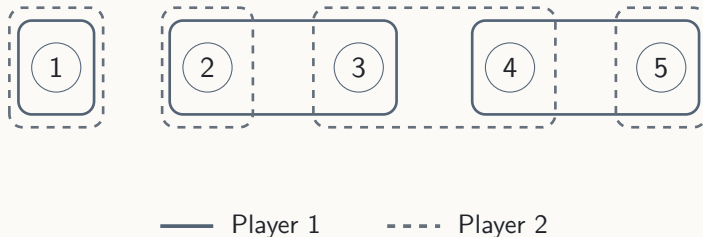
What do we know about one another?

One world; two information partitions

Partitional information for every player.

$$\mathcal{P}_1 = \{\{1\}, \{2, 3\}, \{4, 5\}\},$$

$$\mathcal{P}_2 = \{\{1\}, \{2\}, \{3, 4\}, \{5\}\}.$$



Mutual knowledge

$$KE = \bigcap_i K_i E$$



$$E = \{3, 4, 5\}$$

$$K_1 E = \{4, 5\}, \quad K_2 E = \{3, 4, 5\}$$

$$KE = \{4, 5\}$$

Everyone knows that everyone knows

KE everyone knows E

$K^2E = K(KE)$ everyone knows that everyone knows E

$$K^{m+1}E = K(K^mE), \quad m \geq 1$$

Knowledge can disappear under iteration



$$E \quad \{3, 4, 5\}$$

$$KE \quad \{4, 5\}$$

$$K^2E \quad \{5\}$$

$$K^3E \quad \emptyset$$

At state 5, which level fails first?

Follow the possibilities

An edge joins states that a player cannot distinguish.

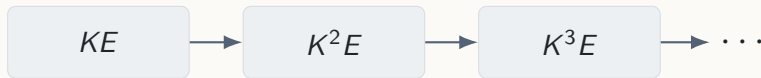


$$5 \xrightarrow{1} 4 \xrightarrow{2} 3 \xrightarrow{1} 2 \notin E$$

$$P(\omega) = \bigcup_i P_i(\omega)$$

Common knowledge: all orders

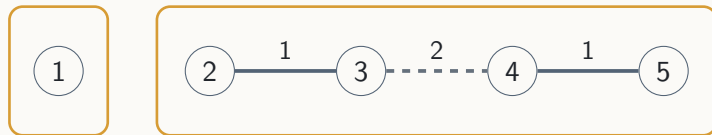
$$\text{CK } E = \bigcap_{m \geq 1} K^m E$$



Every finite chain of reasoning.

Common knowledge: connected components

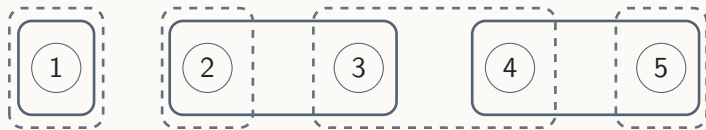
$CP(\omega)$: all states reachable from ω .



$$CP(5) = \{2, 3, 4, 5\}$$

$$\omega \in CK E \iff CP(\omega) \subseteq E$$

The common-knowledge partition

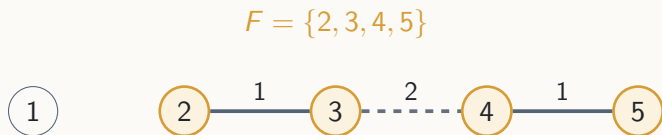


Finest partition coarser than every $\mathcal{P}_i$

$$\{\{1\}, \{2, 3, 4, 5\}\}$$

Each common-knowledge cell is a union of cells of every player.

An event that survives every order

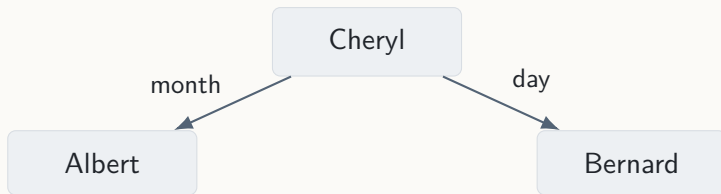


$$K_1 F = K_2 F = F$$

$$KF = K^2 F = \dots = CK F = F$$

Cheryl's birthday

	14	15	16	17	18	19
May		●	●			●
June				●	●	
July	●		●			
August	●	●		●		



Three public statements

Albert

I do not know. I know that Bernard does not know.

Bernard

I did not know. Now I know.

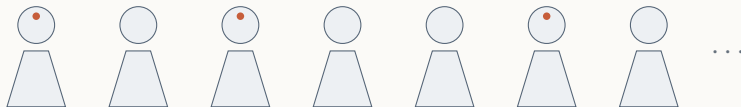
Albert

Now I know too.

Which date remains?

What does a public announcement add?

100 monks. Each sees everyone else's forehead, never their own.



“At least one of you has a red spot.”

Leave that evening as soon as you know you have a spot.

Departures are public; the rules and reasoning are common knowledge.

First departures: evening 7. How many leave?

03

Agreement

Common priors; different observations

From possibilities to probabilities

Partitional information from now on.

Prior

$$p_i \in \Delta(\Omega)$$

Before observing the information cell.

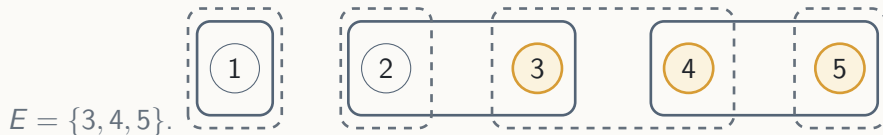
Posterior

$$p_i(E \mid P_i(\omega))$$

$$p_i(E \mid P_i(\omega)) = \frac{p_i(E \cap P_i(\omega))}{p_i(P_i(\omega))}$$

A common prior need not give common posteriors

Common prior p , with $p(\omega) > 0$ at every state. Here: five equally likely states;

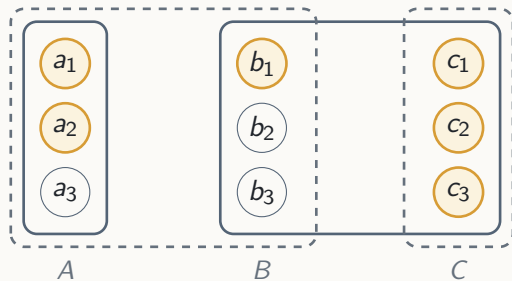


$$q_i(\omega) := p(E \mid P_i(\omega))$$

ω	1	2	3	4	5
$q_1(\omega)$	0	$1/2$	$1/2$	1	1
$q_2(\omega)$	0	0	1	1	1

Can the posteriors be known to one another?

Nine equally likely states. Gold states form E .



$$A = \{a_1, a_2, a_3\},$$

$$B = \{b_1, b_2, b_3\},$$

$$C = \{c_1, c_2, c_3\}.$$

$$\mathcal{P}_1 = \{A, B \cup C\},$$

$$\mathcal{P}_2 = \{A \cup B, C\}.$$

— Player 1 - - - - Player 2

Mutual knowledge permits disagreement

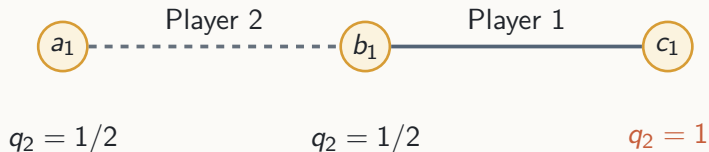
State lies in	A	B	C
q_1	$2/3$	$2/3$	$2/3$
q_2	$1/2$	$1/2$	1

$$D := \{q_1 = 2/3, q_2 = 1/2\} = A \cup B$$

$$K_1 D = A, \quad K_2 D = A \cup B$$

$$KD = A \quad \text{but} \quad CK D = \emptyset$$

The path that breaks common knowledge



$$\text{CP}(a_1) = \Omega$$

The value of Player 2's posterior changes within the component.

Aumann's agreement theorem (1976)

A common prior.

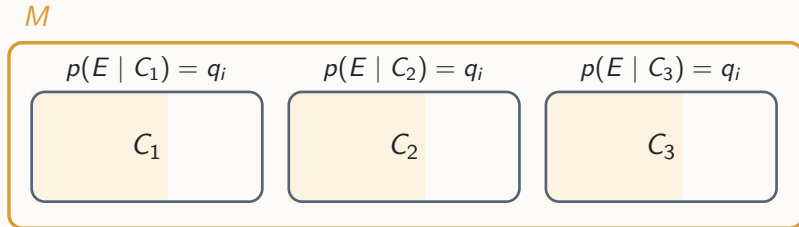
The values of all players' posteriors on E
are common knowledge at ω .

$$q_i(\omega) = q_j(\omega) \quad \text{for every } i, j \in I$$

R. J. Aumann, "Agreeing to Disagree", 1976.

Proof: work inside one common-knowledge cell

$$M := \text{CP}(\omega), \quad q_i := p(E \mid P_i(\omega))$$



M is a union of Player i 's information cells.

Common knowledge makes the posterior constant throughout M .

Proof: average over those cells

$C_1, \dots, C_r$ partition M into Player i 's information cells.

$$\begin{aligned} p(E \mid M) &= \sum_{\ell=1}^r p(C_\ell \mid M) p(E \mid C_\ell) \\ &= \sum_{\ell=1}^r p(C_\ell \mid M) q_i \\ &= q_i. \end{aligned}$$

The same $p(E \mid M)$ for every player.

Can disagreement itself be common knowledge?

$$\Omega = \{a, b\}, \quad p(a) = p(b) = 1/2, \quad E = \{a\}.$$



	a	b
q_1	1	0
q_2	1/2	1/2

$$\text{CK}\{q_1 \neq q_2\} = \Omega$$

A stronger disagreement?

Can it be common knowledge that

$$q_1 > q_2 ?$$

Return to the common-knowledge cell.

From knowledge to strategic reasoning

Possibilities

$$P_i(\omega)$$

Common knowledge

$$CP(\omega)$$

Agreement

$$q_i = p(E \mid CP(\omega))$$

Next: what follows from common knowledge of rationality?

Knowledge and its representations

Aumann (1999), *Interactive epistemology I: Knowledge*.

Possibility correspondences, knowledge operators and their axioms.

Common knowledge and agreement

Aumann (1976), *Agreeing to Disagree*.

Common priors and common knowledge of posterior beliefs.

Further reading

Fagin, Halpern, Moses and Vardi (1995), *Reasoning About Knowledge*.

A systematic treatment of individual and common knowledge. Rubinstein (1998), *Modeling Bounded Rationality*. MIT Press.

Conclusion

Knowledge

Non-delusion and introspection give a partition representation;
knowledge operators describe the same information through events.

Common knowledge

Iterating mutual knowledge identifies the common-knowledge cell:
all states reachable through players' information cells.

Agreement

With a common prior, posterior beliefs that are common knowledge
must coincide.

Next: what follows from common knowledge of rationality?

Check the model: answers

	Non-delusion	Introspection
$P_i(ab) = \{a_0, \dots, a_9\}$	Yes	Yes
$P_i(ab) = \{ba\}$	No	No
$P_i(G) = P_i(B) = \{G\}$	No	Yes

In the second row, consider $a \neq b$. In the third, $\Omega = \{G, B\}$.

Cheryl's birthday: successive restrictions

Public statement	Remaining dates
Albert: Bernard cannot know	July 14, 16; August 14, 15, 17
Bernard: now I know	July 16; August 15, 17
Albert: now I know	July 16

The monks: induction on the number of spots

Sick monks	Each sick monk sees	First departures
1	0 others	evening 1
2	1 other	evening 2
n	$n - 1$ others	evening n

Evening 7: exactly 7 monks leave.

Common knowledge of a strict ordering is impossible

Suppose $q_1(\omega') > q_2(\omega')$ throughout $M = \text{CP}(\omega)$.

$$\mathbb{E}_p[q_1 \mid M] > \mathbb{E}_p[q_2 \mid M]$$

$$\mathbb{E}_p[q_1 \mid M] = p(E \mid M) = \mathbb{E}_p[q_2 \mid M]$$

Contradiction.

One more agreement exercise

Seven equally likely states; $E = \{3, 5, 6\}$.

$$\mathcal{P}_1 = \{\{1, 2, 3\}, \{4, 5\}, \{6, 7\}\},$$

$$\mathcal{P}_2 = \{\{1\}, \{2\}, \{3\}, \{4, 5, 6, 7\}\}.$$

1. Compute both players' posteriors.
2. Find the common-knowledge partition.
3. Where are both posterior values common knowledge?